

Technical Overview of the Palm Computing® Platform

Maurice Sharp
Platform Development Services Manager
Palm Computing

3Com
More connected.

Agenda

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- Palm Computing® Platform Overview
- Palm OS® Software Development
- Conduit Development
- Next Steps

Platform Overview: Architectures

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HotSync® Technology





Palm OS® Software

Handheld Applications

Hardware Add-ons

Accessories

HotSync Manager

Conduits

Platform Overview: Hardware

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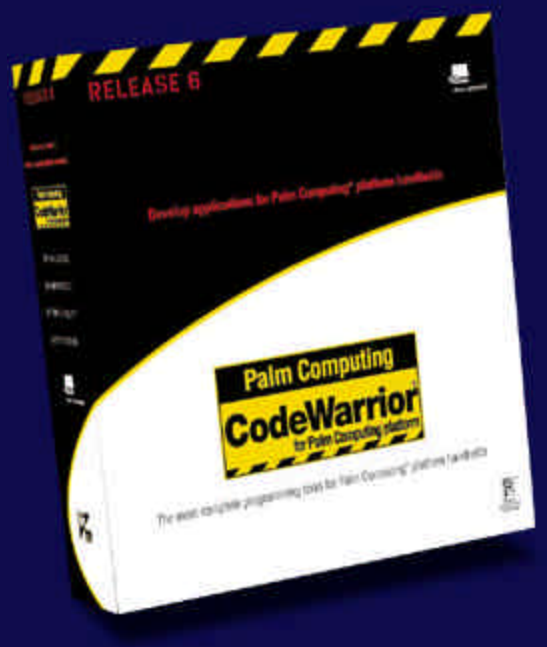
Palm OS® Software Development

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- Tools
  - C/C++
  - Palm OS Emulator
  - Forms tools
- Palm OS® Software

CodeWarrior for Palm OS® Software

- Metrowerks CodeWarrior IDE
- Constructor (resource editor)
- PalmRez (resource compiler)
- C/C++/Assembly
- Comes with
  - Palm OS SDK
  - Palm OS Emulator



Palm OS® Emulator

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- Emulates device hardware
- Runs on Windows/Mac/Linux
- Requires a ROM image
- Sanity checking
- Performance measurement
- Source freely available
- Debugging


Palm OS Emulator
メモ帳
すべて
1. オンラインヒント上級...
2. 基本操作
3. 文字を入力する3通り...
新規
Debugging Options
Low-Memory Access
System Globals Access
Screen Access
Hardware Register Access
MemMgr Data Structure
Storage Heap Access
MemMgr Semaphore
Low Stack Access
Free Chunk Access
Unlocked Chunk Access
Uninitialized Stack Access
Uninitialized Chunk Access
Stack Almost Overflow
Cancel
OK

Gnu PalmPilot SDK

- Freeware
- Available for Windows and Unix
- Command-line using modified version of GCC
- Includes PilRC, a resource compiler
- Several GUI tools exist

Forms Tools

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- Satellite Forms
  - Puma Technology
  - Visual Basic-like scripting language
  - VBX Conduit
  - $795/$995

www.pumatech.com

- Pendragon Forms
  - Pendragon Software
  - Some scripting
  - Includes a Conduit
  - $149

www.pendragon-software.com

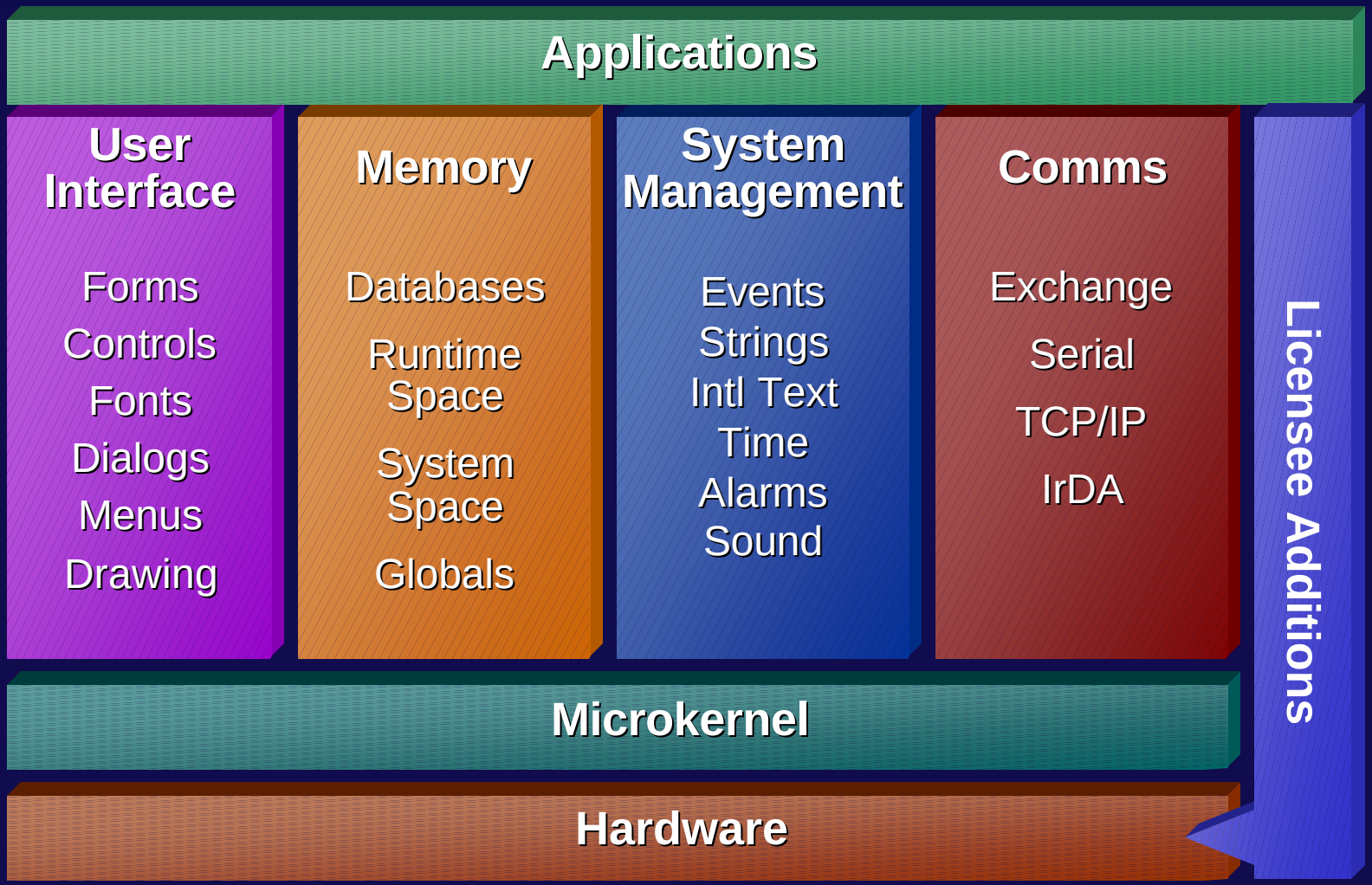
Palm OS® Software

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- Designed for handheld use
- Small, fast, elegant
  - 3.0 is 1.4 megs (including built-in applications!)
  - Max runtime memory use is 56K (including TCP/IP)
- Easy to learn API
- Easy to use API

Palm OS® Architecture

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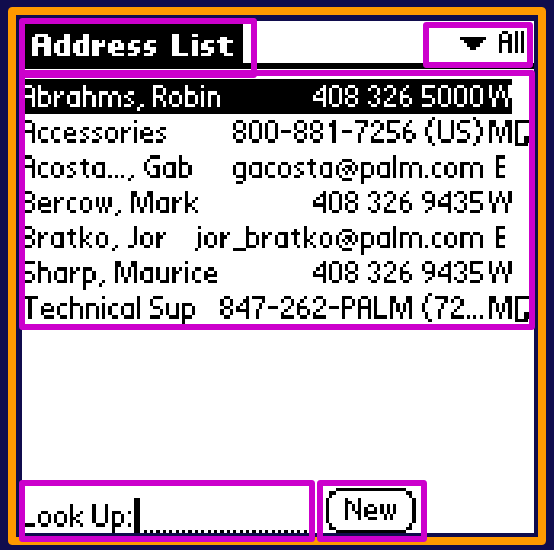
Applications

- One application runs at a time
- Can sub-launch other applications
- Event driven
  - Main event loop for application
  - Dispatches events to handlers
  - Events:
    - UI actions (pen down, button pressed)
    - System notifications
    - Application specific

Forms and Controls

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Form Form Objects



Buttons



Checkboxes



Selectors



Labels and Fields



Pickers

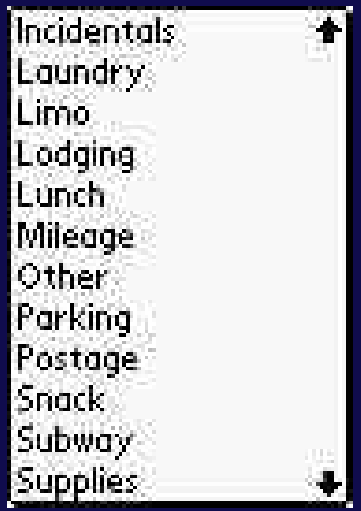


More UI Elements

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Tables & Lists

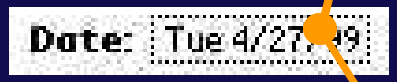




Dialogs/Alerts



Custom Pickers



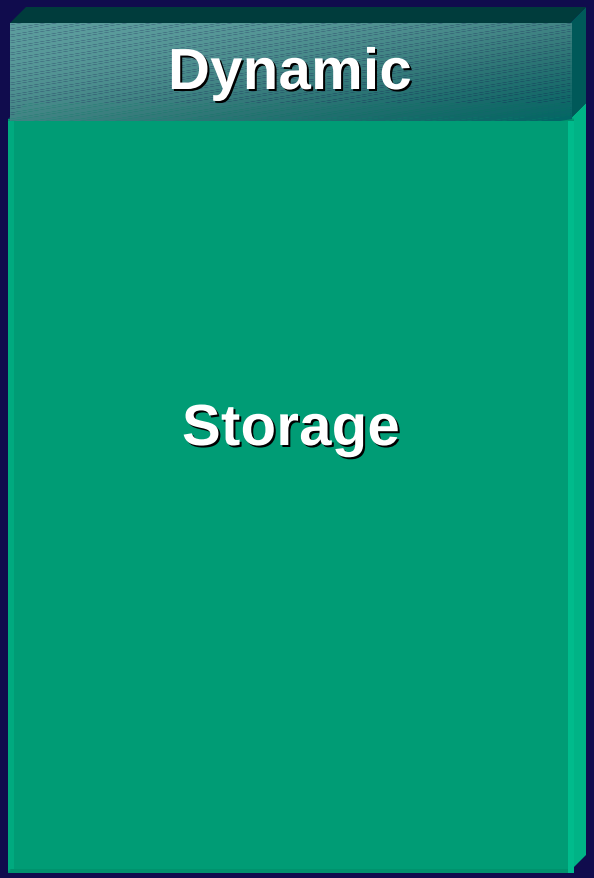


Menus



Memory

- All RAM based
  - Fast access
- Dynamic heap
  - Execution-based storage
    - System
    - Application
  - “Reboot” clears
- Storage heap
  - Write Protected
  - Applications
  - Persistent data
  - “Reboot” does not clear



Memory: Dynamic Heap

- Between 32 and 96K
- Global variables
- Execution Stack
- Dynamic Allocations
  - Nonrelocatable (Pointers)
  - Relocatable (Handles)

Memory: Storage Heap

- Databases
  - The data for your application
  - Store data as records or resources
- File Streaming API
- <stdio.h> API

Records and Resources

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- For application defined data
- Record attributes
  - Dirty/category/secret
- Record access
  - Unique Id or Index
- Resource access
  - Type and Id
- Maximum size 64K

System Management

- Support for program functions
  - String library
  - International Text manager
  - Floating point
  - Time & Date (& Alarms)
  - Events

Communications

- Exchange Manager
  - High level, no comms required
  - Data exchange
- New Serial Manager
  - Manages virtual serial interfaces
  - Serial over RS232, IR…
- TCP/IP
  - PPP
  - Wireless
- IrDA

Conduit Development

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- Tools
- HotSync Architecture
- Conduits
- Server Synchronization

Tools

- Conduit CDK 3.0 Windows
  - Visual C++ 5.0 or Borland C++ Builder
  - CDK 4.0 will support VC++ 6.0
- Conduit CDK Java for Windows
  - Visual Café
  - Visual J++ with JVM 1.1.3
- Conduit CDK Macintosh (2.1)
  - Designed for CodeWarrior Pro 3

Palm OS® Architecture

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Conduit

HotSync® Manager

Task Sequencing
Logging
User Manager
Comm Monitor

Platform Specifics

Conduit Base Classes
InstallAide
CondManager
File Linking

SyncManager API

Palm Database Access
Palm Record Access

PC / Mac / Solaris

Hardware

Target OS File System API

Conduits

- May perform different activities
  - Simple backup
  - Uploading and/or downloading
  - Mirror-image synchronization
  - Offline Transaction Processing (OFTP)
- Standard Backup Conduit

How Conduit Is Packaged

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- DLL on Windows
- Shared Library on Mac OS
- Solaris still in development

Creating a Conduit

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- Implement standard set of entry points
  - Conduit name and version
  - Open conduit
  - Configuration
- Implement Logging
- Implement Synchronization
  - SyncManager API
  - Conduit Base Classes

Server Synchronization

- HotSync® technology is Palm™ device to single desktop
- Third party solutions today
  - Sybase
  - Oracle
  - Puma
  - …
  - Scout
  - ASL Connect
  - AvantGo

Next Steps

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- Stop by the booth
- Attend the other sessions
- Visit the website
  - http://www.palm.com/devzone/
- Become a Solution Provider
  - http://spp.palm.com/
- Programming Books
- Take the Training


Q&A


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